

DUKE BIKE E-COMMERCE DATABASE MANAGEMENT SYSTEM

SELLER AND INVENTORY MANAGEMENT

1. Database Design

Seller and Inventory tables maintain seller details, product stock, availability and update information.

```
CREATE TABLE Seller (  
    Seller_ID NUMBER PRIMARY KEY,  
    Seller_Name VARCHAR2(100) NOT NULL,  
    Contact_Number VARCHAR2(15),  
    Email VARCHAR2(100),  
    Address VARCHAR2(200)  
);  
  
CREATE TABLE Inventory (  
    Inventory_ID NUMBER PRIMARY KEY,  
    Seller_ID NUMBER NOT NULL,  
    Product_ID NUMBER NOT NULL,  
    Stock_Quantity NUMBER DEFAULT 0,  
    Stock_Status VARCHAR2(20) DEFAULT 'In Stock',  
    Last_Updated DATE NOT NULL,  
    CONSTRAINT fk_seller FOREIGN KEY (Seller_ID)  
        REFERENCES Seller(Seller_ID),  
    CONSTRAINT fk_product FOREIGN KEY (Product_ID)  
        REFERENCES Product(Product_ID)  
);
```

Output

SELLER_ID	SELLER_NAME	CONTACT_NUMBER	EMAIL	ADDRESS
301	Chennai Duke Motors	9876543210	chennai@dukebike.com	Chennai
302	Royal Bike Hub	9876543211	royal@bikehub.com	Chennai
303	KTM Power Motors	9876543212	ktm@powermotors.com	Chennai
304	Speed Wheels	9876543213	speed@wheels.com	Chennai
305	Elite Bike World	9876543214	elite@bikeworld.com	Chennai

2. Seller Product Information

```
SELECT s.Seller_Name,
       p.Product_Name,
       i.Stock_Quantity,
       i.Stock_Status
FROM Seller s
JOIN Inventory i ON s.Seller_ID = i.Seller_ID
JOIN Product p ON i.Product_ID = p.Product_ID
ORDER BY s.Seller_Name;
```

SQL Developer-style Output

SELLER_NAME	PRODUCT_NAME	STOCK_QUANTITY	STOCK_STATUS
Chennai Duke Motors	Duke 390	25	In Stock
Chennai Duke Motors	Duke 250	10	In Stock
Elite Bike World	Duke 390	12	In Stock
KTM Power Motors	Duke 250	8	In Stock
Royal Bike Hub	Duke 250	15	In Stock
Speed Wheels	Duke 200	20	In Stock

3. Product Availability

```
SELECT p.Product_ID, p.Product_Name,
       i.Stock_Quantity, i.Stock_Status
FROM Product p
JOIN Inventory i ON p.Product_ID = i.Product_ID
WHERE i.Stock_Status = 'In Stock';
```

Available Products — Output

PRODUCT_ID	PRODUCT_NAME	STOCK_QUANTITY	STOCK_STATUS
201	Duke 390	25	In Stock
202	Duke 250	10	In Stock
204	Duke 250	15	In Stock
205	Duke 390	8	In Stock
207	Duke 200	20	In Stock
208	Duke 390	5	In Stock
210	Duke 250	12	In Stock

4. Unavailable Products

```
SELECT p.Product_ID, p.Product_Name,
       i.Stock_Quantity, i.Stock_Status
FROM Product p
JOIN Inventory i ON p.Product_ID = i.Product_ID
WHERE i.Stock_Status = 'Out of Stock';
```

Output

PRODUCT_ID	PRODUCT_NAME	STOCK_QUANTITY	STOCK_STATUS
203	Duke 200	0	Out of Stock
206	Duke 200	0	Out of Stock
209	Duke 390	0	Out of Stock

5. Inventory Update

```
UPDATE Inventory
SET Stock_Quantity = 20,
    Stock_Status = 'In Stock',
    Last_Updated = TO_DATE('17-09-2026', 'DD-MM-YYYY')
WHERE Inventory_ID = 305;

UPDATE Inventory
SET Stock_Quantity = 12,
    Stock_Status = 'In Stock',
    Last_Updated = TO_DATE('17-09-2026', 'DD-MM-YYYY')
WHERE Inventory_ID = 310;

COMMIT;
```

Output

INVENTORY_ID	STOCK_QUANTITY	STOCK_STATUS	LAST_UPDATED
305	20	In Stock	17-09-2026
310	12	In Stock	17-09-2026

6. Inventory Status Report

```
SELECT s.Seller_Name,  
       SUM(i.Stock_Quantity) AS Total_Stock  
FROM Seller s  
JOIN Inventory i ON s.Seller_ID = i.Seller_ID  
GROUP BY s.Seller_Name  
ORDER BY s.Seller_Name;
```

Output

SELLER_NAME	TOTAL_STOCK
Chennai Duke Motors	35
Elite Bike World	12
KTM Power Motors	28
Royal Bike Hub	15
Speed Wheels	25

```
SELECT Stock_Status, COUNT(*) AS Product_Count  
FROM Inventory  
GROUP BY Stock_Status;
```

Output

STOCK_STATUS	PRODUCT_COUNT
In Stock	7
Out of Stock	3

Note: Product_ID and Product_Name values in these sample outputs should match the Product table already created in the Duke Bike database.

Conclusion

The Seller and Inventory Management module maintains seller details, stock information, availability status, inventory updates and seller-wise inventory reports.